Automated Decision · Explainability Report

IrisNet classifier · Test case #1 · Generated 2026-10-14 00:52 UTC

DECISION

Virginica · 74.3% **Actual: Virginica (correct)**

1. Identification

Model	IrisNet (4-5-3 MLP), 43 parameters
Model fingerprint	**6a8caca14b1c7521**
Test accuracy	93.3%
Case analysed	Test sample #1
Alternatives	Setosa 0.0%, Versicolor 25.7%

2. Input

Sepal length	6.1 cm
Sepal width	3.0 cm
Petal length	4.9 cm
Petal width	1.8 cm

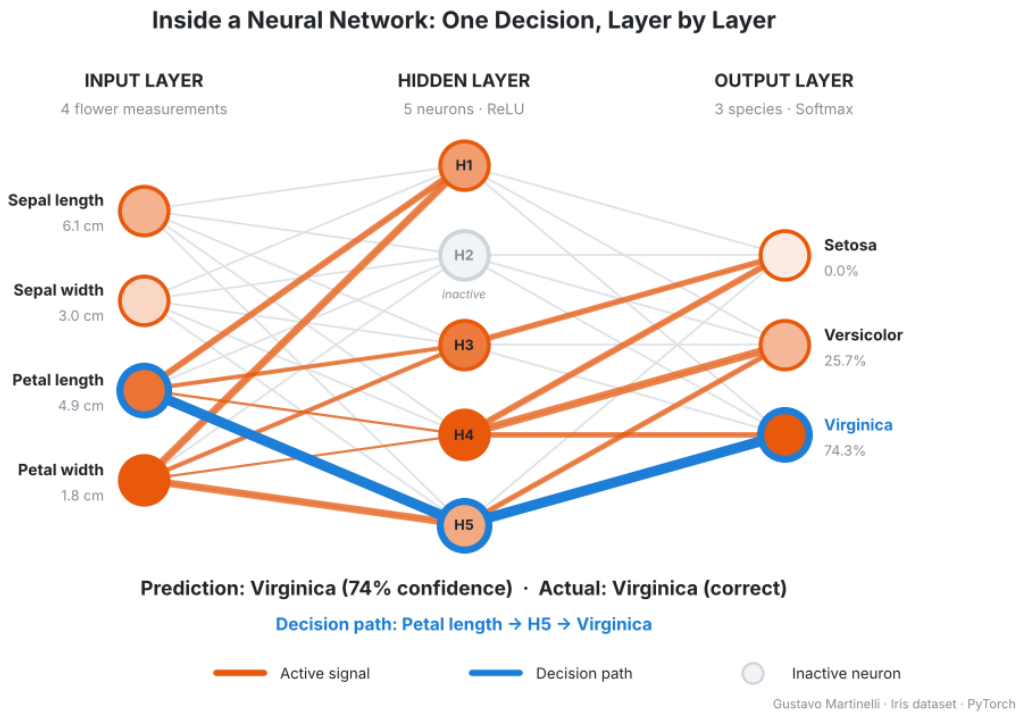
3. Main reasons (influence on the predicted class)

Supports = pushed toward this species; opposes = pushed away. The rest of the decision comes from the model's learned starting point.

1. Petal length	48%	supports
2. Petal width	46%	supports
3. Sepal length	4%	supports
4. Sepal width	2%	supports

4. Internal decision path

Dominant route	**Petal length → H5 → Virginica**
Inactive neurons	H2



5. Method and limitations

- Influence = exact gradient × input decomposition for this input, measured relative to the average flower in the training data.
- The decision path highlights the strongest route; every active neuron contributes to the final result.
- This report explains one decision only, not the model's overall behaviour.
- Human review is recommended before relying on this decision.